Experiment No.1

EXPERIMENT NO – 1

Aim of the Experiment:

The goal is to understand how network cables are made and how to create them. You'll learn how to make straight-through and cross-wired cables using the right tools.

What You Need:

- **RJ45 Connectors** (The plugs at the ends of the cables)
- **Clamping Tool** (The tool used to attach the connector to the cable)
- **Twisted Pair Cable** (The type of cable commonly used for network connections)

Steps to Follow:

1. **Prepare the Cable:**
    - Start by removing about 2 inches of the outer plastic from the end of the cable. Be careful not to damage the wires inside.
    - Separate the wires carefully and make sure they are not tangled.
2. **Arrange the Wires:**
    - After separating the wires, arrange them in a specific order. There are two main ways to arrange them:
        - **Straight-through Cable**: This is the most common way of wiring, used for connecting a computer to a switch or a router.
        - **Cross-wired Cable**: This is used to connect similar devices, like connecting two computers directly.
    - Follow the wiring diagram closely to ensure each wire goes in the correct slot.
3. **Attach the Plug:**
    - Once the wires are in the correct order, carefully insert them into the RJ45 connector.

- Use the clamping tool to press the connector onto the cable. This makes sure the wires are securely connected to the plug.

4. **Test the Cable:**

   - After finishing the cable, you can test it with a cable tester (if available) to make sure all connections are working correctly.

Diagram shows you how to prepare Cross wired connection

RJ45 Pin # (END 1)	Wire Color	Diagram End #1	RJ45 Pin # (END 2)	Wire Color	Diagram End #2
1	White/Orange		1	White/Green	
2	Orange		2	Green	
3	White/Green		3	White/Orange	
4	Blue		4	White/Brown	
5	White/Blue		5	Brown	
6	Green		6	Orange	
7	White/Brown		7	Blue	
8	Brown		8	White/Blue	

Experiment No.2

EXPERIMENT NO – 2

Aim:

To understand and study the following network devices:

- Repeater
- Hub
- Switch
- Bridge
- Router
- Gateway

1. Repeater

A repeater is a network device used to extend the range of a network. When a signal becomes weak while travelling a long distance, the repeater picks it up, strengthens it, and sends it forward so it can cover more distance without getting lost. It has only two ports, so it can only connect two devices. It works at the Physical Layer (Layer 1) of the OSI model.

2. Hub

A hub is a network device used to connect multiple devices together in a network. When it receives data from one device, it sends that data to all connected devices — even if it was meant for only one. Because of this, it creates extra unnecessary traffic and makes the network slower. It works at the Physical Layer (Layer 1) of the OSI model.

3. Switch

A switch is a network device used to connect multiple devices, just like a hub — but it is smarter. It sends data only to the device it is meant for, not to everyone. It does this by remembering the address (MAC address) of each connected device, which reduces unnecessary traffic and makes the network faster. It works at the Data Link Layer (Layer 2) of the OSI model.

4. Bridge

A bridge is a network device used to connect two separate network segments and control the traffic between them. It checks each incoming data packet and decides whether to pass it forward or block it based on the device address. This helps manage traffic and improve overall network performance. It works at the Data Link Layer (Layer 2) of the OSI model.

5. Router

A router is a network device used to send data from one network to another. It checks the IP address of every data packet and decides which path is best to reach the destination. If multiple routers are connected, they share information with each other to always find the best route. It works at the Network Layer (Layer 3) of the OSI model.

6. Gateway

A gateway is a network device that connects two networks which use different protocols (different communication rules). It acts like a translator — it converts the data from one format to another so both networks can understand each other. It may include protocol translators, signal converters, and rate converters. It works at multiple layers of the OSI model.

Experiment No.3

Aim:

Study of Network IP addresses, Classification of IP addresses, Subnetting, and Supernetting.

IP Address

An IP address is a unique address assigned to every device on a network, used to identify it and send data to the correct destination. It consists of 4 numbers separated by dots, for example 192.168.1.1.

Classification of IP Addresses

Class	Address Range	Supports
Class A	1.0.0.0 – 126.255.255.255	126 networks, 16 million hosts each
Class B	128.0.0.0 – 191.255.255.255	16,000 networks, 65,000 hosts each
Class C	192.0.0.0 – 223.255.255.255	2 million networks, 254 hosts each
Class D	224.0.0.0 – 239.255.255.255	Reserved for multicast groups
Class E	240.0.0.0 – 255.255.255.255	Reserved for future use

Subnetting

Subnetting is used when we want to divide a large network into smaller parts. Each smaller part is called a subnet. This helps in organising and managing the network better.

Supernetting

Supernetting is the opposite of subnetting. It is used when we want to combine several smaller networks into one large network. This reduces the number of networks a router needs to manage.

Experiment No.4

Aim -

Connect computers in a Local Area Network (LAN) and share the internet connection between them.

On the Host Computer -

The host computer is the one that has the internet connection and shares it with others.
Steps

1. Log in as Administrator or Owner
2. Click Start → Control Panel
3. Click Network and Internet Connections
4. Click Network Connections
5. Right-click the connection you use for internet (e.g. modem) → click Properties
6. Go to the Advanced tab
7. Under Internet Connection Sharing, tick "Allow other network users to connect through this computer's internet connection"
8. A message will appear asking confirmation — click Yes
9. Click OK

On the Client Computer -

The client computer is the one that wants to use the shared internet from the host.
Steps

1. Log in as Administrator or Owner
2. Click Start → Control Panel
3. Click Network and Internet Connections
4. Click Network Connections
5. Right-click Local Area Connection → Properties
6. Click General tab → Internet Protocol (TCP/IP) → Properties
7. Select "Obtain an IP address automatically" and click OK
8. Click OK in the Local Area Connection Properties box

Experiment No.5

Experiment No. 5

Aim:

To study basic network commands and basic network configuration commands.

Apparatus (Software):

- Command Prompt
- Packet Tracer

Procedure:

In this experiment, students study commonly used basic networking commands such as **ping, tracert,** and network configuration commands such as how to switch modes, configure interfaces, and save settings to memory.

---

1) Ping –

Ping is used to check if another device on the network is reachable. It sends a small test message to the device — if a reply comes back, the device is active and working. If there is no reply, it means something is wrong with the connection.

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2) Tracert –

Tracert stands for Trace Route. It is used to show the full path a data packet takes from your computer to a destination device. It lists every router the packet passes through on its way, one by one, until it reaches the destination — or fails. Along with the path, it also tells you how long each step (hop) takes. This helps in finding out where exactly a connection is slow or breaking.

---

3) Pathping –

Pathping is a combination of ping and tracert. It shows the full path a packet takes and also gives extra details like how many packets were lost and how much delay there was at each step. This makes it more useful when you want to find the exact problem point in a network.

---

4) Configure –

The configure command is used on a network device to enter configuration mode, where you can make changes to its settings. Once inside, you can set up interfaces, assign IP addresses, and change other network settings. You can also shorten commands — for example, show can be typed as sh, as long as it is unique from other commands.

---

5) Show ip route –

This command is used to display the routing table of a router. The routing table is a list of all the network paths the router knows about. It shows which networks can be reached and through which path. The router uses this table to decide where to send each data packet it receives.

Configuration Files -

Whenever you make changes to network configuration settings, you must save them — otherwise they will be lost when the router restarts or loses power.

There are two types of configuration files:

• **Running configuration** — current active settings, stored in temporary memory (lost on restart)
• **Startup configuration** — saved settings that load when the router starts, stored in permanent memory

Result:

Thus, basic network commands and network configuration commands were successfully studied.

In Computer execution -

Watch this video -
https://www.awesomescreenshot.com/video/51507959?key=86fbd950a3c7fd5f94a899e3015596c4

Open terminal from cisco tracer app, set up the connections (system, connection and ip address) , type following commands -

1) ping 10.10.10.1
2) tracert 10.10.10.1
3) nslookup
4) ipconfig

NOTE - make sure to wait after entering each command to see the output!!

Experiment No.6

Experiment No. 6

Aim:

To perform **initial configuration of a Cisco Catalyst 2960 switch**.

Background / Preparation:

In this experiment, the basic settings of a Cisco switch are configured, such as:

- Host name
- Console password
- VTY password
- Privileged mode password and secret
- IP address on VLAN 1
- Default gateway

Procedure:

Step 1: Configure Host Name

```
enable
configure terminal
hostname CustomerSwitch
```

Step 2: Configure Privileged Mode Password

```
enable password cisco
enable secret cisco123
```

Step 3: Configure Console Password

```
line console 0
password cisco
```

```
login
exit
```

Step 4: Configure VTY Password

```
line vty 0 15
password cisco
login
exit
```

Step 5: Configure IP Address on VLAN 1

```
interface vlan 1
ip address 10.10.10.3 255.0.0.0
no shutdown
Click enter once more
exit
```

Step 6: Configure Default Gateway

```
ip default-gateway 10.10.10.2
```

Step 7: Verify Configuration

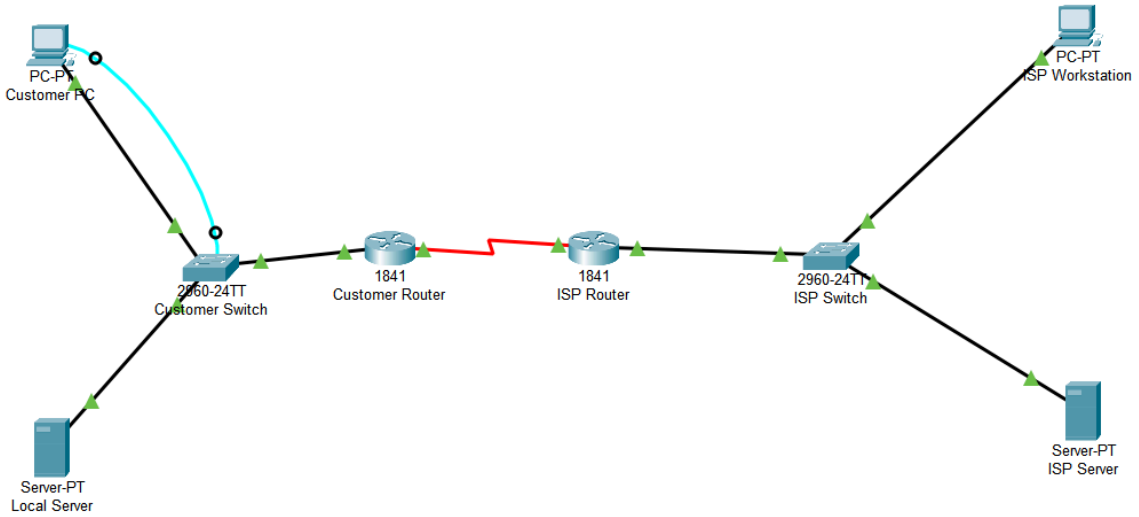
```
end
```

Click enter once more

```
ping 30.30.30.1
```

Result:

The Cisco Catalyst 2960 switch was **successfully configured** with basic settings and connectivity was verified.



WATCH THIS VIDEO -
https://www.awesomescreenshot.com/video/51514249?key=253c164db69934b67675d51c73b0dca8

Config address-

Customer PC IP: `10.10.10.1`
Customer PC Gateway: `10.10.10.2`
Customer Router 0/0: `10.10.10.2`

Customer Router Serial: `20.20.20.1`
ISP Router Serial: `20.20.20.2`

ISP Workstation/Server IP: `30.30.30.1`
ISP Workstation Gateway: `30.30.30.2`
ISP Router 0/0: `30.30.30.2`

Customer Router RIP: Add `10.0.0.0` and Add `20.0.0.0`
ISP Router RIP: Add `20.0.0.0` and Add `30.0.0.0`

Experiment No.7

Experiment 7 — Performing an Initial Router Configuration

Aim
To perform an initial configuration on a Cisco 1841 router using the Cisco IOS CLI.

Objectives

- Configure the router host name
- Configure passwords
- Configure banner message
- Verify the router configuration

Background / Preparation
In this activity, the Cisco IOS CLI is used to set up a router for the first time. This includes setting a host name, passwords, a banner message and other basic settings.

Steps -

Step 1 — Configure the Router Host Name
```
enable
configure terminal
hostname CustomerRouter
```

Step 2 — Configure Privileged Mode Password and Secret
```
enable password cisco
enable secret cisco123
```

Step 3 — Configure Console Password
```
line console 0
password cisco
login
exit
```

Step 4 — Configure VTY Password
```
line vty 0 15
password cisco
login
exit
```

Step 5 — Configure Encryption, Banner and Save
```
service password-encryption
banner motd $Authorized Access Only!$
no ip domain-lookup
```

```
interface fastethernet0/0
ip address 10.10.10.2 255.0.0.0
no shutdown
end
copy run start
```

Step 6 — Verify the Configuration

a. Log out of the router by typing exit twice:

```
exit
```

b. The banner message "Authorized Access Only!" will appear on screen.

c. Log back in by entering the console password:

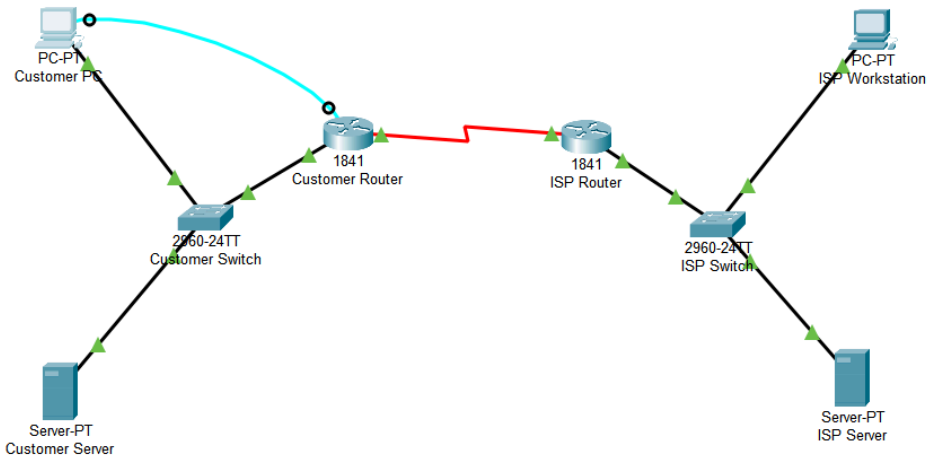
```
Password: cisco
```

d. Enter privileged EXEC mode:

```
enable
Password: cisco123
```

e. Verify the configuration using:

```
show running-config
```



Device	Interface	IP Address	Subnet Mask	Gateway
Customer PC	FastEthernet0	10.10.10.1	255.0.0.0	10.10.10.2
Customer Server	FastEthernet0	10.10.10.3	255.0.0.0	10.10.10.2
Customer Router	FastEthernet0/0	10.10.10.2	255.0.0.0	N/A
Customer Router	Serial0/0/0	20.20.20.1	255.0.0.0	N/A
ISP Router	Serial0/0/0	20.20.20.2	255.0.0.0	N/A
ISP Router	FastEthernet0/0	30.30.30.2	255.0.0.0	N/A
ISP Workstation	FastEthernet0	30.30.30.1	255.0.0.0	30.30.30.2
ISP Server	FastEthernet0	30.30.30.3	255.0.0.0	30.30.30.2

Experiment No.8

Experiment 8 — Configuring and Troubleshooting a Switched Network

Aim

To configure basic settings on a Cisco Catalyst 2960 switch.

Objectives

- Configure hostname and VLAN 1 IP
- Configure passwords and banner
- Configure port security
- Save configuration

Configuration Commands

1. Configure Customer Switch (CLI)

```
enable
configure terminal
hostname CustomerSwitch
interface vlan 1
ip address 10.10.10.4 255.0.0.0
no shutdown
exit
ip default-gateway 10.10.10.2
enable secret class
line console 0
password cisco
login
exit
line vty 0 15
password cisco
login
exit
banner motd $Authorized Access Only$
interface fastethernet0/18
switchport port-security
end
copy run start
```

2. Configure Customer Router (CLI)

```
enable
configure terminal
interface fastethernet0/0
ip address 10.10.10.2 255.0.0.0
no shutdown
end
copy run start
```

3. Configure Customer PC1 (Desktop → IP Configuration)

- IP Address: 10.10.10.1
- Subnet Mask: 255.0.0.0
- Default Gateway: 10.10.10.2

4. Configure Customer PC2 (Desktop → IP Configuration)

- IP Address: 10.10.10.3
- Subnet Mask: 255.0.0.0
- Default Gateway: 10.10.10.2

Verification

On Customer Switch CLI:

```
show running-config
```

On Customer PC1 Command Prompt:

```
ipconfig
ping 10.10.10.3
```

Experiment No.9

Experiment 9 — Examining WAN Connection

Aim
To examine WAN connection and use various show commands to view and monitor network connectivity.

Objective
Show commands are powerful tools used for troubleshooting and monitoring networks. They give a clear picture of how the network is communicating and transferring data at a given time.

Background / Preparation
The network topology is built using Frame Relay. To test and verify the network connectivity, various show commands are used.

Required file: Examining WAN connections.pka

Steps

Step 1 — Examine the configuration of Branch1 and Branch2 (router)

a. To view the current configuration of the router.
show running-config

b. To view the status of all interfaces.
show interfaces

c. To get a quick summary of all interfaces.
show ip interface brief

d. To check the Frame Relay PVC status.
show frame-relay pvc

e. To check the LMI status of the Frame Relay circuit.
show frame-relay lmi

Step 2 — Examine the configuration of Main(router)

a. To view the current configuration.
show running-config

b. To view the status of all interfaces of the Main router.
show interfaces

c. To get a quick summary of all interfaces.
show ip interface brief

d. To verify the Frame Relay PVC status on Main router.
show frame-relay pvc

e. To check the LMI status and verify the Frame Relay circuit.
show frame-relay lmi

Result
The WAN connection was successfully examined. The Frame Relay network topology was verified and the status of all interfaces and circuits was confirmed.

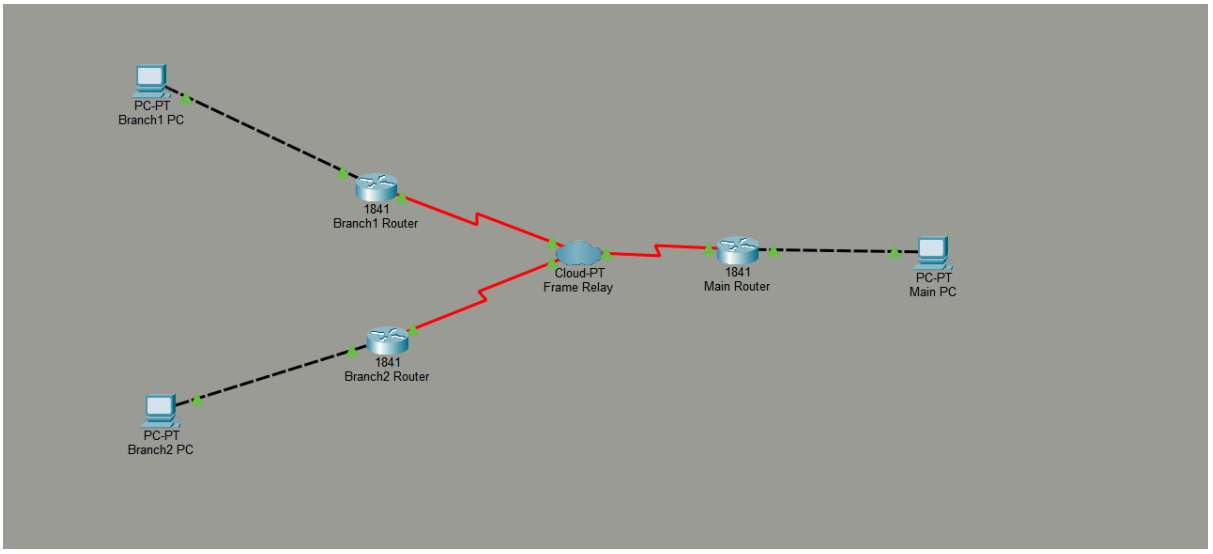
NOTE - frame-relay pvc & frame-relay lmi , is not working because we didnt configure cloud
If you want to make frame-relay lmi command work, use this-
enable
configure terminal
interface serial0/0/0
encapsulation frame-relay
end

Then show frame-relay lmi

Config address-

Device	IP Address	Subnet Mask	Gateway
Branch1 PC	10.10.10.1	255.0.0.0	10.10.10.2
Branch1 Router (Fa0/0)	10.10.10.2	255.0.0.0	N/A
Branch1 Router (Serial)	20.20.20.1	255.0.0.0	N/A
Branch2 PC	10.10.10.3	255.0.0.0	10.10.10.4
Branch2 Router (Fa0/0)	10.10.10.4	255.0.0.0	N/A
Branch2 Router (Serial)	20.20.20.3	255.0.0.0	N/A
Main PC	30.30.30.1	255.0.0.0	30.30.30.2
Main Router (Fa0/0)	30.30.30.2	255.0.0.0	N/A
Main Router (Serial)	20.20.20.2	255.0.0.0	N/A

Watch this video -
https://www.awesomescreenshot.com/video/51583476?key=df6c06e1d83395294021b207025e9aec



Experiment No.10

Experiment — Interpreting Ping and Traceroute Output

Aim

To test end-to-end connectivity using ping and traceroute.

Objectives

- Distinguish between successful and unsuccessful ping attempts
- Distinguish between successful and unsuccessful traceroute attempts

Background / Preparation

In this activity, you will test connectivity using ping and traceroute. Make sure the network is converged (all link lights green) before beginning.

Steps

Step 1 — Test Connectivity Using Ping

Click **Customer PC** → **Desktop** → **Command Prompt**

Type:

text Copy Download

```text
ping www.cisco.com
```

- **Successful ping:** Receives replies from the destination
- **Unsuccessful ping:** Shows "Request timed out"

Step 2 — Test Connectivity Using Traceroute

From the same Command Prompt, type:

text Copy Download

```text
tracert www.cisco.com
```

- **Successful traceroute:** Shows each hop (router) along the path
- **Unsuccessful traceroute:** Shows "Request timed out" after a certain hop

---

Result

Ping and traceroute commands were successfully executed to test network connectivity.

Step 3 — Configure IP Addresses (Using Config Tab, No CLI)

PC0:

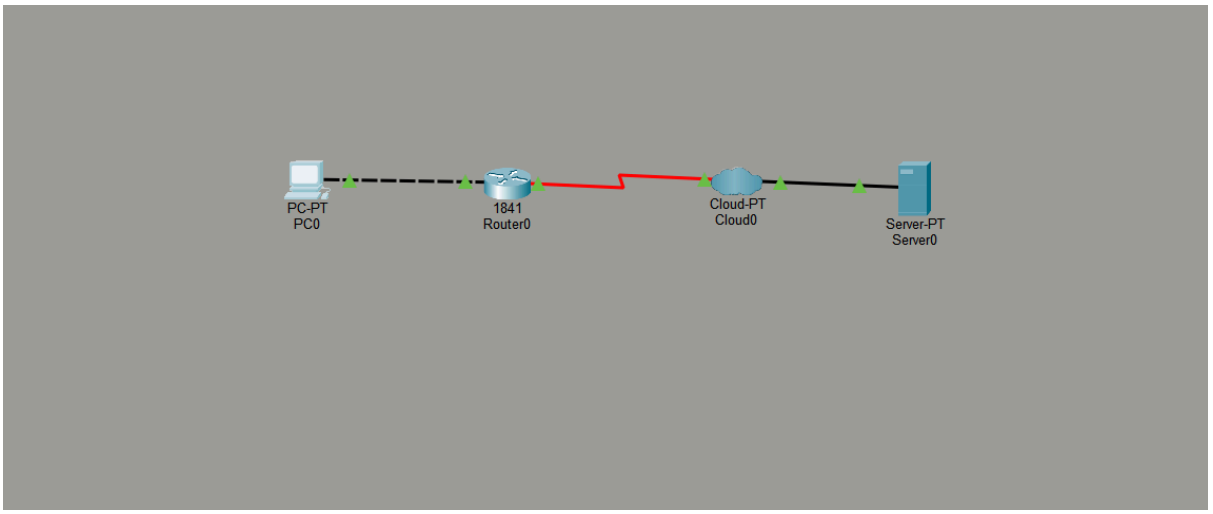
- IP: 10.10.10.1
- Mask: 255.0.0.0
- Gateway: 10.10.10.2

Router0 (Config tab):

- FastEthernet0/0: IP 10.10.10.2, Mask 255.0.0.0, Port Status ON
- Serial0/0/0: IP 20.20.20.1, Mask 255.0.0.0, Clock Rate 64000, Port Status ON

Server0:

- IP: 30.30.30.3
- Mask: 255.0.0.0
- Gateway: 30.30.30.2



Device	IP	Gateway
PC0	10.10.10.1	10.10.10.2
Router0 (Fa0/0)	10.10.10.2	–
Router0 (Serial)	20.20.20.1	–
Server0	30.30.30.3	30.30.30.2

What to Tell Teacher:

> *"In Packet Tracer, we cannot ping the real www.cisco.com. So we created a server (Server0) with IP 30.30.30.3 to act as the Cisco server. This simulates pinging a real web server."*

Experiment No.12

Experiment — Implementing an IP Addressing Scheme

Aim
To implement an IP addressing scheme and test connectivity between devices using ping commands.

Objectives
- Assign IP addresses to devices
- Verify IP configuration using ipconfig
- Test connectivity using ping

Background / Preparation
Two PCs are connected directly or through a switch. Each PC is assigned an IP address. Connectivity is tested using ping.

Steps -

Step 1 — Configure IP Addresses on Devices
On PC0 (Host A):
- IP Address: 10.10.10.1
- Subnet Mask: 255.0.0.0
- Default Gateway: 10.10.10.2

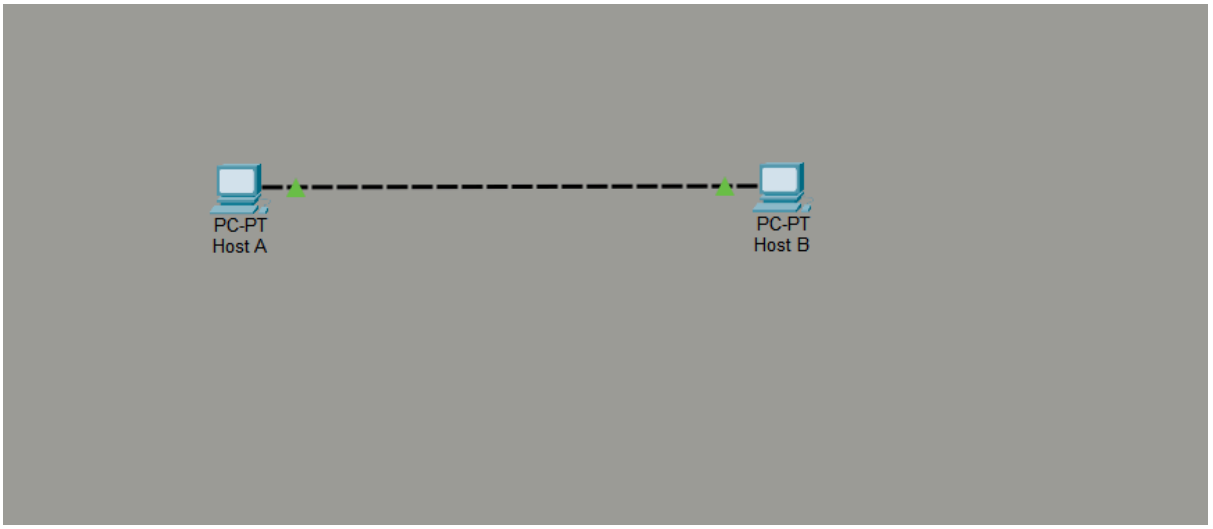
On PC1 (Host B):
- IP Address: 10.10.10.3
- Subnet Mask: 255.0.0.0
- Default Gateway: 10.10.10.2

NOTE - since they are in the same network we have used same address for gateway.

Step 2 — Verify IP Configuration
On PC0 (Host A), open Command Prompt and type:
ipconfig

Step 3 — Test Connectivity Using Ping

From PC0 (Host A), ping Host B:
ping 10.10.10.3


PC-PT
Host A
PC-PT
Host B